

D777

Real Life Applications of Data Structures

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A ticketing app checks each of 50,000 scanned barcodes against a Python list of 2 million valid codes, and the check is slow. The app runs on a cloud VM with 8 vCPUs. What is the best fix?

- A. Load the valid codes into a set and test membership there.
- B. Move to a VM with more vCPUs.
- C. Sort the list, then keep using in.
- D. Convert the list to a tuple.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A library system must find a book's shelf location from its ISBN thousands of times per minute. The library has 14 branches. Which structure fits best?

- A. A dict mapping ISBN to shelf location.
- B. A set of ISBNs.
- C. A list of (ISBN, location) tuples scanned each time.
- D. Fourteen separate lists, one per branch.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A print server must print jobs in the order they arrive. The office has 60 employees and one color printer. A developer adds jobs with `list.append()` and takes the next job with `list.pop()`. What is wrong?

- A. Nothing; `append` and `pop` form a queue.
- B. Use a set to hold the jobs.
- C. `pop()` takes the newest job (LIFO); use `collections.deque` with `popleft()` for FIFO.
- D. Buy a second printer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An editor's undo feature must reverse the most recent edit first. Which structure and operations fit?

- A.** A deque with `popleft()`.
- B.** A priority queue ordered by edit size.
- C.** A stack: append each edit, pop to undo.
- D.** A dict keyed by edit number.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A function finds duplicate customers by comparing every record with every other record using two nested loops over n records. The dataset was cleaned last month. What is the time complexity?

- A.** $O(\log n)$
- B.** $O(n^2)$
- C.** $O(n \log n)$
- D.** $O(n)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

An $O(n^2)$ routine takes 2 seconds on 10,000 records. About how long will it take on 30,000 records?

- A.** 9 seconds
- B.** 54 seconds
- C.** 6 seconds
- D.** 18 seconds

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A developer uses Python's bisect module to search a list of 1 million order IDs that are appended in arrival order, not sorted. Searches return wrong answers. Each ID is 12 digits long. Why?

- A. bisect performs a linear search.
- B. Binary search only works on sorted data.
- C. 12-digit IDs are too long to compare.
- D. bisect fails on lists over 100,000 items.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A 300-person firm stores employee IDs in a binary search tree. Starting empty, it inserts 50, 30, 70, 20, 40. What does an in-order traversal return?

- A. 50, 30, 20, 40, 70
- B. 70, 50, 40, 30, 20
- C. 20, 30, 40, 50, 70
- D. 50, 30, 70, 20, 40

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Keys 1, 2, 3, ..., n are inserted in that order into a plain (unbalanced) binary search tree. What is the worst-case search time?

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n \log n)$
- D. $O(n)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A transit app must find the route with the fewest transfers between stations, where every hop counts as 1. The network has 400 stations and was redesigned in 2024. Which algorithm fits best?

- A. Only Dijkstra’s algorithm works for any shortest path.
- B. Breadth-first search.
- C. Depth-first search.
- D. Sort the 400 stations alphabetically first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An emergency room triage tool must always pull the patient with the highest severity score next, with frequent new arrivals. The hospital has 30 beds. The developer uses Python’s heapq, a min-heap. How should entries be stored?

- A. Use a FIFO queue.
- B. Push severity as is.
- C. Keep a list and sort it after every arrival.
- D. Push (–severity, patient_id) tuples so heappop returns the most severe patient.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Using heapq, a program pushes 7, 3, 9, and 1 onto an empty heap. What does the first heappop return?

- A. 1
- B. 9
- C. 7
- D. 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A ticketing app checks each of 50,000 scanned barcodes against a Python list of 2 million valid codes, and the check is slow. ~~The app runs on a cloud VM with 8 vCPUs.~~ What is the best fix?

Target word: speed up membership **Decides:** in on a list scans every element.

A	Correct	Set membership is $O(1)$ on average.
B	Camouflage	Hardware does not change the $O(n)$ check.
C	Dropped element	in on a list stays linear even when sorted.
D	Confusion set	Tuple membership is also linear.

2. Answer A Concept 2

A library system must find a book's shelf location from its ISBN thousands of times per minute. ~~The library has 14 branches.~~ Which structure fits best?

Target word: structure for lookup **Decides:** Need a value (location) for each key (ISBN).

A	Correct	Average $O(1)$ key-to-value lookup.
B	Dropped element	Answers whether a book exists but stores no location.
C	Adjacent category	Works, but each lookup is $O(n)$.
D	Camouflage	Echoes the branch count; still linear scans.

3. Answer C Concept 3

A print server must print jobs in the order they arrive. ~~The office has 60 employees and one color printer.~~ A developer adds jobs with `list.append()` and takes the next job with `list.pop()`. What is wrong?

Target word: queue discipline **Decides:** `pop()` removes the last item added.

A	Inverted operation	They form a stack.
B	Confusion set	Sets are unordered.
C	Correct	A queue must remove the oldest item first.
D	Camouflage	Hardware does not fix ordering.

4. Answer C Concept 3 · transfer

An editor's undo feature must reverse the most recent edit first. Which structure and operations fit?

Target word: structure — reversed to LIFO **Decides:** Most recent first.

A	Inverted operation	That undoes the oldest edit first.
B	Confusion set	Orders by size, not recency.
C	Correct	LIFO returns the latest edit first.
D	Adjacent category	Can store edits but does not give LIFO order by itself.

5. Answer B Concept 4

A function finds duplicate customers by comparing every record with every other record using two nested loops over n records. ~~The dataset was cleaned last month.~~ What is the time complexity?

Target word: Big-O **Decides:** $n \times n$ comparisons.

A	Confusion set	Describes halving, as in binary search.
B	Correct	Each of n records is compared with about n others.
C	Adjacent category	Describes efficient sorting, not all-pairs comparison.
D	Dropped element	Drops the inner loop.

6. Answer D Concept 4 · transfer

An $O(n^2)$ routine takes 2 seconds on 10,000 records. About how long will it take on 30,000 records?

Target word: solve for time **Decides:** n triples, so time grows by $3^2 = 9$.

A	Dropped element	The growth factor without the 2-second base.
B	Confusion set	Cubic scaling ($3^3 \times 2$).
C	Adjacent category	Linear scaling.
D	Correct	$2 \times 9 = 18$.

7. Answer B Concept 5

A developer uses Python's bisect module to search a list of 1 million order IDs that are appended in arrival order, not sorted. Searches return wrong answers. ~~Each ID is 12 digits long.~~ Why?

Target word: why bisect fails **Decides:** The list is unsorted.

A	Confusion set	It performs binary search.
B	Correct	Halving assumes order; unsorted data breaks it.
C	Camouflage	ID length does not decide correctness.
D	Absolute language	No such limit; it scales as $O(\log n)$.

8. Answer C Concept 6

~~A 300-person firm stores employee IDs in a binary search tree.~~ Starting empty, it inserts 50, 30, 70, 20, 40. What does an in-order traversal return?

Target word: in-order traversal **Decides:** In-order visits left, node, right.

A	Confusion set	Pre-order traversal.
B	Inverted operation	Reverse order (right, node, left).
C	Correct	In-order traversal of a BST yields sorted order.
D	Adjacent category	Level order, which here matches insertion order.

9. Answer D Concept 6 · transfer

Keys 1, 2, 3, ..., n are inserted in that order into a plain (unbalanced) binary search tree. What is the worst-case search time?

Target word: search cost — degenerate case **Decides:** Each key becomes the right child of the last.

A	Confusion set	Hash-table lookup, not tree search.
B	Dropped element	Assumes balance the tree does not have.
C	Adjacent category	Sorting cost, not a single search.
D	Correct	The tree becomes a chain.

10. Answer B Concept 7

A transit app must find the route with the fewest transfers between stations, where every hop counts as 1. ~~The network has 400 stations and was redesigned in 2024.~~ Which algorithm fits best?

Target word: algorithm choice **Decides:** Unweighted graph, fewest edges.

A	Absolute language	Dijkstra works but is unnecessary when all edges weigh 1.
B	Correct	BFS finds the fewest-edge path in an unweighted graph.
C	Confusion set	DFS does not guarantee the shortest path.
D	Camouflage	Station order does not find paths.

11. Answer D Concept 8

An emergency room triage tool must always pull the patient with the highest severity score next, with frequent new arrivals. ~~The hospital has 30 beds.~~ The developer uses Python's heapq, a min-heap. How should entries be stored?

Target word: max-priority with heapq **Decides:** heapq pops the smallest value.

A	Confusion set	Serves by arrival, not severity.
B	Inverted operation	Pops the least severe first.
C	Adjacent category	Works but costs $O(n \log n)$ per insert.
D	Correct	Negating turns the min-heap into max-first order.

12. Answer A Concept 8 · transfer

Using heapq, a program pushes 7, 3, 9, and 1 onto an empty heap. What does the first heappop return?

Target word: heap output **Decides:** Min-heap returns the smallest.

A	Correct	heapq always pops the minimum.
B	Inverted operation	Would be true for a max-heap.
C	Confusion set	First pushed, as a FIFO queue would return.
D	Wrong stem number	The second value pushed; not the minimum.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	List vs. set membership	x in list is $O(n)$; x in set is $O(1)$ on average because sets are hash-based.	Sorting a list or switching to a tuple and expecting fast in.	Many membership checks against a large collection.
2	Dictionaries for key-value lookup	A dict maps keys to values with average $O(1)$ lookup.	Scanning a list of pairs, or using a set that stores no value.	“Look up X from Y” thousands of times.
3	Stack vs. queue	A stack is LIFO (append / pop); a queue is FIFO (deque append / popleft).	Using list.pop() for first-in-first-out work.	“In the order they arrive” vs. “most recent first.”
4	Big-O of nested loops	A loop over n inside a loop over n is $O(n^2)$; doubling n quadruples the time.	Scaling $O(n^2)$ work linearly.	Comparing every item with every other item.
5	Binary search needs sorted data	Binary search (including bisect) is $O(\log n)$ and correct only on sorted input.	Running bisect on unsorted data.	Data appended in arrival order, then bisected.
6	Binary search trees	In-order traversal of a BST gives sorted order; inserting sorted keys into an unbalanced BST makes it a chain with $O(n)$ search.	Assuming every BST search is $O(\log n)$.	An insertion sequence and a traversal question.
7	Shortest paths in graphs	BFS finds fewest-edge paths in unweighted graphs; Dijkstra handles non-negative weights; DFS does not guarantee shortest paths.	Using DFS for shortest path, or insisting on Dijkstra for equal weights.	“Fewest hops” or “each step counts as 1.”
8	Heaps and priority queues	Python’s heapq is a min-heap; store negated priorities to pop the largest first.	Pushing priorities directly and expecting the maximum.	“Highest priority next” with heapq.

Course scope (public description): Real Life Applications of Data Structures covers basic to advanced topics in data structures and their associated algorithms with a focus on real world examples and applications. This course focuses on problem-solving techniques for designing efficient and maintainable software solutions using Python.